

Performance Testing

Project Title

Virtual Agent–Driven SLA Breach Awareness & Justification System

Introduction

Performance Testing was conducted to evaluate the efficiency, responsiveness, reliability, and stability of the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. The primary objective was to verify that the application performs efficiently under normal operating conditions while accurately monitoring SLA status, processing workflows, generating AI-powered responses, and updating dashboards in real time.

The testing focused on validating the performance of major system components, including incident management, SLA monitoring, Virtual Agent interactions, workflow automation, AI integration, and reporting.

Objectives

The objectives of Performance Testing were to:

- Verify that the application responds quickly to user requests.
- Ensure accurate real-time SLA monitoring.
- Validate workflow automation performance.
- Measure response time of the Virtual Agent.
- Verify Google Gemini AI response generation.
- Ensure reliable database operations.
- Confirm dashboard updates occur without delay.
- Evaluate system stability during continuous usage.

Test Environment

Parameter	Configuration
Platform	ServiceNow & Flask Application
Backend	Python (Flask)
Database	SQLite
Browser	Google Chrome
Operating System	Windows 11

Parameter	Configuration
AI Service	Google Gemini AI
Test Users	Support Agent, Team Lead, Administrator
Network	Stable Internet Connection

Test Cases

Test Case 1 – User Login

Objective: Verify authentication performance.

Parameter	Result
Expected Result	User logs in successfully.
Actual Result	Login completed successfully.
Status	Passed

Test Case 2 – Incident Creation

Objective: Verify incident creation speed.

Parameter	Result
Expected Result	Incident created and stored successfully.
Actual Result	Incident created successfully.
Status	Passed

Test Case 3 – SLA Monitoring

Objective: Validate real-time SLA monitoring.

Parameter	Result
Expected Result	SLA timer updates continuously.
Actual Result	SLA monitored accurately in real time.
Status	Passed

Test Case 4 – SLA Risk Classification

Objective: Verify automatic SLA classification.

Parameter	Result
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Expected Result Incident classified as SAFE, AT RISK, or BREACHED.

Actual Result	Classification performed successfully.
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Status	Passed
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Test Case 5 – Virtual Agent Response

Objective: Measure Virtual Agent responsiveness.

Parameter	Result
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Expected Result Virtual Agent provides timely responses.

Actual Result	Responses generated successfully.
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Status	Passed
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Test Case 6 – AI Justification Generation

Objective: Validate Google Gemini AI integration.

Parameter	Result
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Expected Result AI generates SLA breach explanation and recommendations.

Actual Result	AI response generated successfully.
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Status	Passed
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Test Case 7 – Workflow Automation

Objective: Verify Flow Designer execution.

Parameter	Result
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Expected Result Automated workflows execute successfully.

Actual Result	Workflow triggered and completed successfully.
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Status	Passed
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Test Case 8 – Dashboard Update

Objective: Verify dashboard refresh performance.

Parameter	Result
Expected Result	Dashboard displays updated SLA statistics.
Actual Result	Dashboard updated successfully.
Status	Passed

Test Case 9 – Report Generation

Objective: Validate report generation.

Parameter	Result
Expected Result	Reports generated accurately.
Actual Result	Reports generated successfully.
Status	Passed

Performance Metrics

Metric	Expected Outcome	Observed Outcome
User Login	Fast authentication	Achieved
Incident Creation	Quick record creation	Achieved
SLA Monitoring	Continuous monitoring	Achieved
SLA Classification	Accurate classification	Achieved
Virtual Agent	Responsive interaction	Achieved
AI Justification	Timely AI response	Achieved
Workflow Automation	Successful execution	Achieved
Dashboard Update	Real-time updates	Achieved
Report Generation	Accurate reporting	Achieved
System Stability	Stable during testing	Achieved

Observations

- User authentication operated successfully.
- Incidents were created and updated without errors.

- SLA timers monitored incidents continuously.
 - SLA classification was performed accurately.
 - Virtual Agent interactions were responsive.
 - Google Gemini AI generated meaningful SLA breach explanations.
 - Flow Designer workflows executed successfully.
 - Dashboards reflected real-time SLA information.
 - Reports displayed accurate operational data.
 - No critical performance issues were observed during testing.
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Performance Evaluation

The application demonstrated:

- Fast user authentication.
 - Efficient incident processing.
 - Accurate SLA monitoring.
 - Reliable AI-assisted recommendations.
 - Stable workflow automation.
 - Secure database transactions.
 - Responsive dashboard visualization.
 - Consistent application performance under normal operating conditions.
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Recommendations

The following improvements may further enhance system performance:

- Optimize database queries for large-scale deployments.
 - Introduce caching for frequently accessed dashboard data.
 - Improve AI response optimization for high request volumes.
 - Implement asynchronous processing for notification services.
 - Add continuous performance monitoring.
 - Enhance scalability for enterprise-level incident volumes.
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Conclusion

Performance Testing confirmed that the **Virtual Agent–Driven SLA Breach Awareness & Justification System** performs efficiently and reliably across all major functional areas. User authentication, incident management, SLA monitoring, workflow automation, Virtual Agent interactions, Google Gemini AI integration, dashboard visualization, and reporting operated successfully during testing. The system demonstrated stable performance, accurate processing, and responsive behavior, making it suitable for deployment as an intelligent IT Service Management solution.